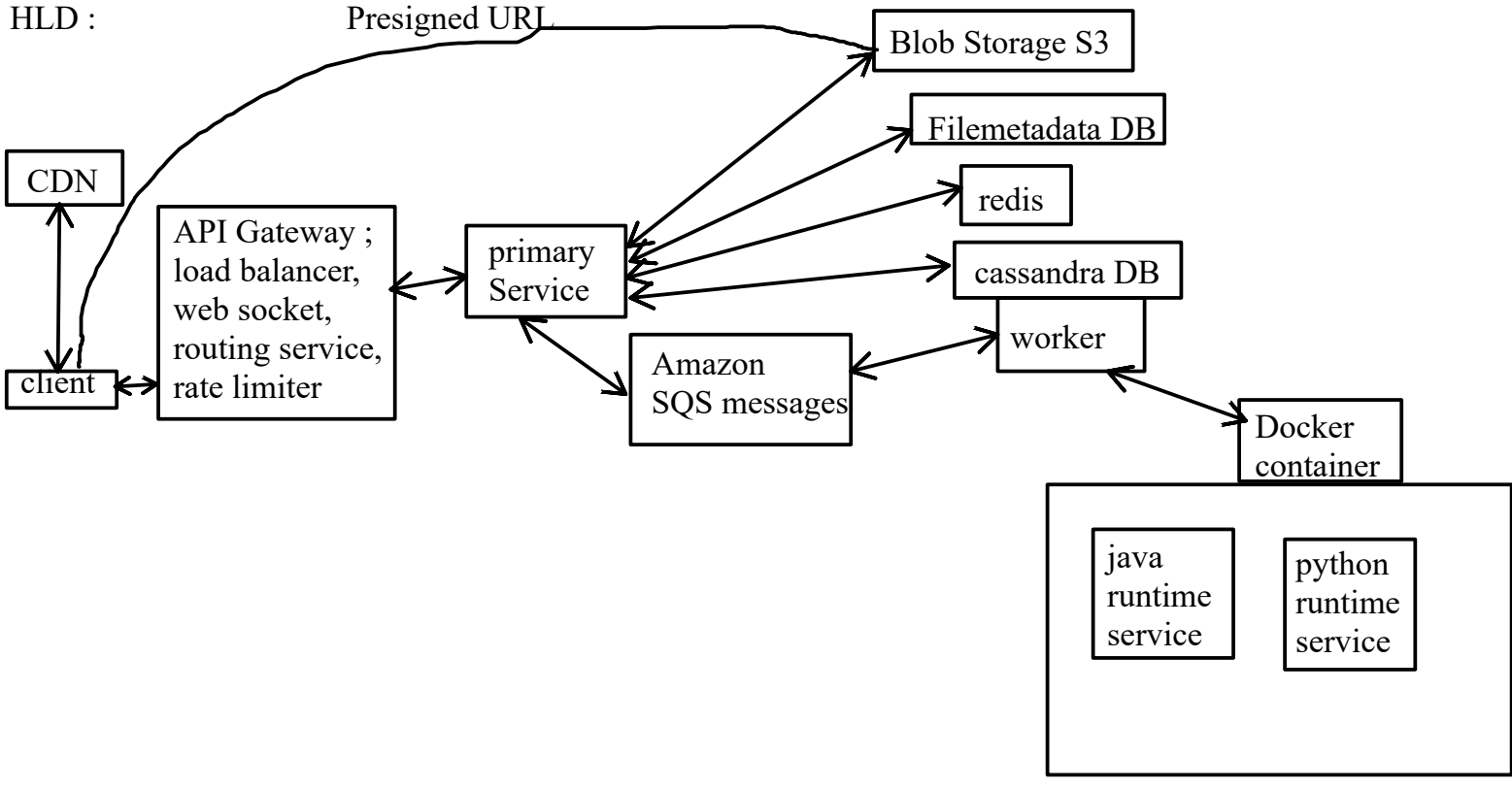
Leetcode

Functional requirements:
user should be able to solve & submit solution based on his preference,
user should be able to search for any problem based on his preference,
user should be able to participate in coding contest based on his preference

Non-functional requirements:
Data-partitioing,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability ,
low latency,
high throughput,
CAP theorem(Here we need to prioritise Strong consistency for rank predictions during coding contest & for search scenario we need to priotise 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
problem,
contest,
history,
solution

API :
POST:submit/solution
{
userID,
{filemetadata}
}
GET:/search/{problemID}

HLD :